CCNA LEARNING SERIES

Day 6: Inter-VLAN Routing

VLAN communication • Router-on-a-Stick • Layer 3 switching • Cisco commands

Learning objective
Understand why inter-VLAN routing is required, learn two common methods, and practise basic Cisco IOS configuration.

Topics: Inter-VLAN routing basics; Router-on-a-Stick; Layer 3 switch; configuration; verification; troubleshooting; interview Q&A; practice questions.

1. What is Inter-VLAN Routing?

A VLAN (Virtual Local Area Network) divides a switched network into separate logical broadcast domains. Devices in different VLANs normally belong to different IP subnets. **Inter-VLAN routing** is the process of forwarding traffic between those VLANs using a router or Layer 3 switch.

Why is it needed?
Example: VLAN 10 is HR (192.168.10.0/24) and VLAN 20 is IT (192.168.20.0/24). A PC in VLAN 10 cannot reach a PC in VLAN 20 through Layer 2 switching alone. A Layer 3 device routes between the subnets.

Device	Example
PC 1 – VLAN 10	192.168.10.10/24
VLAN 10 gateway	192.168.10.1
PC 2 – VLAN 20	192.168.20.10/24
VLAN 20 gateway	192.168.20.1

Each VLAN/subnet needs a suitable default gateway. A host sends traffic for a different subnet to its default gateway.

2. Common Inter-VLAN Routing Methods

Method	How it works
Legacy routing	A router uses a separate physical interface for each VLAN.
Router-on-a-Stick	One router physical interface uses multiple subinterfaces; the switch-router link is an 802.1Q trunk.
Layer 3 switch	A multilayer switch routes between VLANs using Switched Virtual Interfaces (SVIs).

3. Router-on-a-Stick

A router's single physical interface is divided into logical subinterfaces. Each subinterface is assigned an IP address and VLAN tag. The switch port connected to the router must be a trunk, and the VLAN IDs must match.

4. Configuration Example

Assumptions: Switch S1 has VLAN 10 and VLAN 20. S1 GigabitEthernet0/1 connects to router R1 GigabitEthernet0/0. PC ports are already assigned to the correct VLANs.

Switch configuration (S1)

```
enable
configure terminal
vlan 10
name HR
vlan 20
name IT
interface gigabitEthernet 0/1
switchport mode trunk
no shutdown
end
```

Router configuration (R1)

```
enable
configure terminal
interface gigabitEthernet 0/0
no shutdown
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
end
```

PC settings

PC	IP address	Gateway
VLAN 10 PC	192.168.10.10/24	192.168.10.1
VLAN 20 PC	192.168.20.10/24	192.168.20.1

Test from PC 1: ping 192.168.20.10. A successful reply indicates connectivity, assuming no ACL or host firewall blocks the traffic.

5. Inter-VLAN Routing with a Layer 3 Switch

A Layer 3 switch performs Layer 2 switching and Layer 3 routing. It commonly uses an SVI for each VLAN. The SVI IP address acts as the default gateway for hosts in that VLAN.

Example configuration

```
enable
configure terminal
ip routing
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

VLANs must exist. SVIs generally need an active VLAN with at least one operational access or trunk port in that VLAN for the SVI line protocol to come up. Requirements vary by platform.

6. Verification Commands

Command	Purpose
show vlan brief	Check VLANs and access-port membership.
show interfaces trunk	Check trunk ports and allowed VLANs.
show ip interface brief	Check interface and IP status.
show running-config	Review active configuration.
show ip route	Check the routing table.
ping <destination-IP>	Test IP connectivity.

7. Common Troubleshooting Points

• VLAN is missing or the PC is in the wrong VLAN.
• Switch-to-router link is not operating as a trunk.
• VLAN is not allowed on the trunk.
• Router subinterface has the wrong 802.1Q VLAN ID.
• IP address or default gateway is incorrect.
• Router interface or SVI is down.
• An ACL or host firewall blocks traffic.

8. Interview Questions and Answers

Q1. What is inter-VLAN routing?

Routing traffic between different VLANs using a router or Layer 3 switch.

Q2. Why is it required?

Different VLANs are separate Layer 2 broadcast domains and normally use different IP subnets.

Q3. What is Router-on-a-Stick?

One physical router interface uses multiple subinterfaces to route between VLANs.

Q4. What is a subinterface?

A logical interface created under a physical router interface.

Q5. What is IEEE 802.1Q?

A standard used to tag Ethernet frames with VLAN information on trunk links.

Q6. What is a trunk port?

A switch port that carries traffic for multiple VLANs.

Q7. What is an SVI?

A Switched Virtual Interface is a logical Layer 3 interface associated with a VLAN.

Q8. What does ip routing do on a multilayer switch?

It enables IPv4 routing on platforms that require this command.

Q9. Which command checks trunk ports?

show interfaces trunk.

Q10. Which command displays VLANs?

show vlan brief.

Q11. What is a default gateway?

The Layer 3 address a host uses to send traffic outside its local IP subnet.

Q12. Name two inter-VLAN routing methods.

Router-on-a-Stick and Layer 3 switch routing.

9. Practice Questions

Try these before checking the answers.

1. Which device is needed to route between VLANs?
2. Which standard tags VLAN traffic on a trunk?
3. What is a router subinterface?
4. Which command checks the VLAN list on a Cisco switch?
5. Which command checks trunk interfaces?
6. What is the VLAN 10 default gateway in the example?
7. Which command enables IPv4 routing on many Layer 3 switches?
8. Give two reasons a PC in VLAN 10 may fail to ping a PC in VLAN 20.

Practice Answers

1. Router or Layer 3 switch.
2. IEEE 802.1Q.
3. A logical interface under a physical router interface, configured for a VLAN.
4. show vlan brief.
5. show interfaces trunk.
6. 192.168.10.1.
7. ip routing.
8. Incorrect gateway, missing VLAN, trunk issue, wrong VLAN tag or interface down.

Quick Revision

Term	Key point
Inter-VLAN routing	Routes traffic between separate VLANs.
Router-on-a-Stick	One router link with multiple VLAN subinterfaces.
802.1Q	VLAN tagging standard for trunk links.
SVI	Layer 3 virtual interface associated with a VLAN.
Default gateway	Next hop for traffic outside the local subnet.
show vlan brief	Displays VLANs and access-port membership.
show interfaces trunk	Displays trunk information.

Study tip: Build two VLANs in Packet Tracer, connect a router to the switch using a trunk, configure router subinterfaces and test using ping.

End of Day 6 | Next topic: Static Routing

Note: Interface names, command availability and configuration details vary by Cisco platform and IOS version. Verify commands in your lab device.